
Engineering Mathematics – 1

Tutorial Solutions

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Note:

This document contains Tutorials 1 and 2 with short answers, followed by detailed solutions to selected problems. Try solving the questions yourself before looking at the solutions, as they are most useful for checking your work and understanding.

The notes are updated regularly, and any feedback, corrections, or suggestions for improvement are always welcome. If you spot an issue, please mention the page and relevant section, problem, figure, or equation number to help locate it quickly.

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Tutorial 1: Rolle's Theorem and Lagrange's Mean Value Theorem

Questions marked ► [Problem](#) are solved in full in Sections 1–2; click the link to go to the solution.

1. Verify Rolle's theorem for the following functions.

(a) $f(x) = x^2(1-x)^2$ on $[0, 1]$. ► [Problem 1.1](#) [Ans. $c = \frac{1}{2}$]

(b) $f(x) = \frac{\sin x}{e^x}$ on $[0, \pi]$. [Ans. $c = \frac{\pi}{4}$]

(c) $f(x) = \tan x$ on $[0, \pi]$. [Ans. not applicable: $\tan x$ is discontinuous at $x = \frac{\pi}{2}$]

(d) $f(x) = e^x \sin x$ on $[0, \pi]$. ► [Problem 1.2](#) [Ans. $c = \frac{3\pi}{4}$]

(e) $f(x) = \log\left(\frac{x^2+9}{10x}\right)$ on $[1, 9]$. ► [Problem 1.3](#) [Ans. $c = 3$]

(f) $f(x) = \sqrt{(x-4)(5-x)}$ on $[4, 5]$. [Ans. $c = \frac{9}{2}$]

(g) $f(x) = \sqrt[3]{(x-2)^2}$ on $[1, 3]$. ► [Problem 1.4](#)
[Ans. not applicable: not differentiable at $x = 2$]

2. Apply Rolle's theorem to the function $f(x) = \sin x \sqrt{\cos x}$ on $\left[0, \frac{\pi}{2}\right]$, and hence show that $0 \leq f(x) \leq \sqrt[4]{\frac{4}{27}}$ for $x \in \left[0, \frac{\pi}{2}\right]$. [Ans. $c = \tan^{-1} \sqrt{2} \approx 0.9553$, $f(c) = \sqrt[4]{4/27} \approx 0.6204$]

3. Verify Lagrange's mean value theorem for the following functions.

(a) $f(x) = \cos^2 x$ on $\left[0, \frac{\pi}{2}\right]$. ► [Problem 2.1](#)

Ans. $c = \frac{1}{2} \sin^{-1}\left(\frac{2}{\pi}\right) \approx 0.3451$ or $c = \frac{1}{2} \left[\pi - \sin^{-1}\left(\frac{2}{\pi}\right)\right] \approx 1.2257$

(b) $f(x) = \sin^{-1} x$ on $[0, 1]$. ► [Problem 2.2](#) [Ans. $c = \sqrt{1 - \frac{4}{\pi^2}} \approx 0.7712$]

(c) $f(x) = 2x^3 - 3x + 1$ on $[-2, 2]$. ► [Problem 2.3](#) [Ans. $c = \pm \frac{2}{\sqrt{3}}$]

(d) $f(x) = (x-4)(x-6)(x-8)$ on $[4, 10]$. [Ans. $c = 8$]

(e) $f(x) = \arcsin x$ on $[0, 1]$ (the same function as (b)). [Ans. $c = \frac{\sqrt{\pi^2-4}}{\pi} \approx 0.7712$]

(f) $f(x) = \log x$ on $[1, e]$. [Ans. $c = e - 1$]

4. Apply Lagrange's mean value theorem to $f(x) = \log x$ to prove that $0 < \frac{1}{\log(1+x)} - \frac{1}{x} < 1$ for $x > 0$. ► [Problem 2.4](#)

5. For all $x > 0$, apply Lagrange's mean value theorem to prove the inequalities

$$(i) \cos x > 1 - \frac{x^2}{2}, \quad (ii) e^{-x} > 1 - x.$$

6. A ball is thrown vertically upward from the ground with an initial speed of 29.4 m/s. Its height above the ground after t seconds is $h(t) = 29.4t - 4.9t^2$ metres. Show that at some instant during the flight the ball is momentarily at rest, and find that instant and the maximum height attained. ► [Problem 1.5](#) [Ans. $c = 3$ s, $h(3) = 44.1$ m]

7. A car enters a toll booth at 8:00 a.m. and exits another booth 180 km away at 10:00 a.m. The entire highway has a posted speed limit of 70 km/h. On arrival the driver is issued a speeding citation based solely on the two timestamps, and disputes it, claiming that the speed never exceeded 70 km/h. Using the mean value theorem, justify the claim of the traffic authorities that the driver was speeding. [Ans. $v(c) = 90$ km/h at some instant c , and $90 > 70$]

8. In an automated chemical batch reactor, an exothermic reaction is monitored over an 8-minute cycle, $t \in [0, 8]$. The temperature, in $^{\circ}\text{C}$, is $T(t) = 50 + 12t - 1.5t^2$. (a) Verify that T satisfies all the conditions of Rolle's theorem on $[0, 8]$. (b) Determine the critical time $c \in (0, 8)$ at which the instantaneous rate of temperature change vanishes, and the maximum temperature. ► Problem 1.6 [Ans. $T(0) = T(8) = 50^{\circ}\text{C}$; $c = 4$ min, $T(4) = 74^{\circ}\text{C}$]

9. A storage tank holds 200 litres of liquid at 9:00 a.m. The inlet and outlet valves are rated so that the volume in the tank can never change at a rate exceeding 12 L/min in either direction. An operator's log records that the tank held 500 litres at 9:20 a.m. Show that this entry must be erroneous.

Ans. $|V(20) - 200| \leq 12 \times 20 = 240$, so $V(20) \leq 440 < 500$

10. A simply supported beam of span $L = 6$ m carries a load varying linearly from zero at the left support A to a maximum intensity w_0 at the right support B . Measuring x from A , the deflection is

$$y(x) = Kx(x^4 - 120x^2 + 3024), \quad K = \frac{w_0}{720EI} > 0.$$

Show that a point of maximum deflection exists strictly inside the span, and locate it.

Ans. $c \approx 3.116$ m $= 0.5193L$, $y(c) \approx 0.00652 \frac{w_0 L^4}{EI}$

11. A simplified cam gives the inlet valve lift over a 240° crank-angle event as $L(\theta) = \frac{\theta(240 - \theta)^2}{204800}$ mm, $0 \leq \theta \leq 240$. Show that the valve attains a maximum lift at which it is momentarily stationary, and determine the maximum lift and the corresponding crank angle.

[Ans. $\theta = 80^{\circ}$, $L(80) = 10$ mm]

12. Suppose that f is continuous on $[-7, 0]$ and differentiable on $(-7, 0)$, with $f(-7) = -3$ and $f'(x) \leq 2$ for all $x \in (-7, 0)$. Determine the largest possible value of $f(0)$. ► Problem 2.5

[Ans. 11]

Tutorial 2: Series; Taylor and Maclaurin Series

Questions marked ► [Problem](#) are solved in full in Sections 3–4; click the link to go to the solution.

Series

1. Test the convergence of the following series by the root test.

$$(a) \sum_{n=1}^{\infty} \left(\frac{2n+3}{3n+2} \right)^n. \quad \text{► Problem 3.1} \quad [\text{Ans. } L = \frac{2}{3} < 1: \text{convergent}]$$

$$(b) \sum_{n=1}^{\infty} \left(1 + \frac{1}{n} \right)^{-n^2}. \quad \text{► Problem 3.2} \quad [\text{Ans. } L = \frac{1}{e} < 1: \text{convergent}]$$

$$(c) \left(\frac{2^2}{1^2} - \frac{2}{1} \right)^{-1} + \left(\frac{3^3}{2^3} - \frac{3}{2} \right)^{-2} + \left(\frac{4^4}{3^4} - \frac{4}{3} \right)^{-3} + \dots \quad [\text{Ans. } L = \frac{1}{e-1} < 1: \text{convergent}]$$

$$(d) \sum_{n=1}^{\infty} \frac{7}{(2n+5)^n}. \quad [\text{Ans. } L = 0 < 1: \text{convergent}]$$

$$(e) \sum_{n=1}^{\infty} \left(\frac{4n+3}{3n-5} \right)^n. \quad [\text{Ans. } L = \frac{4}{3} > 1: \text{divergent}]$$

$$(f) \sum_{n=1}^{\infty} \frac{n^2}{2^n}. \quad [\text{Ans. } L = \frac{1}{2} < 1: \text{convergent}]$$

$$(g) \sum_{n=1}^{\infty} \frac{4^n}{(3n)^n}. \quad [\text{Ans. } L = 0 < 1: \text{convergent}]$$

$$(h) \sum_{n=1}^{\infty} \left(1 + \frac{2}{n} \right)^{n^2}. \quad \text{► Problem 3.3} \quad [\text{Ans. } L = e^2 > 1: \text{divergent}]$$

$$(i) \sum_{n=1}^{\infty} \frac{1}{2^n} \left(1 + \frac{1}{n} \right)^{n^2}. \quad [\text{Ans. } L = \frac{e}{2} > 1: \text{divergent}]$$

$$(j) \sum_{n=2}^{\infty} (\log n)^{-2n}. \quad [\text{Ans. } L = 0 < 1: \text{convergent}]$$

$$(k) \sum_{n=1}^{\infty} \frac{1}{\left(1 + \frac{1}{\sqrt{n}} \right)^{n^{3/2}}}. \quad [\text{Ans. } L = \frac{1}{e} < 1: \text{convergent}]$$

2. Test the convergence of the following series by the ratio test.

$$(a) \sum_{n=1}^{\infty} \frac{2^{n-1}}{3^n + 1}. \quad \text{► Problem 3.4} \quad [\text{Ans. } L = \frac{2}{3} < 1: \text{convergent}]$$

$$(b) \sum_{n=1}^{\infty} \frac{n! 3^n}{n^n}. \quad [\text{Ans. } L = \frac{3}{e} > 1: \text{divergent}]$$

$$(c) \sum_{n=0}^{\infty} \frac{2^n + 5}{3^n}. \quad [\text{Ans. } L = \frac{2}{3} < 1: \text{convergent}]$$

$$(d) \sum_{n=0}^{\infty} \frac{n^3 + a}{2^n + a}, \text{ where } a > 0. \quad \text{► Problem 3.5} \quad [\text{Ans. } L = \frac{1}{2} < 1: \text{convergent}]$$

$$(e) \sum_{n=1}^{\infty} \frac{n^2}{2^n}. \quad [\text{Ans. } L = \frac{1}{2} < 1: \text{convergent}]$$

$$(f) \sum_{n=2}^{\infty} \frac{3^{n+2}}{\log n}. \quad \blacktriangleright \text{Problem 3.6} \quad [\text{Ans. } L = 3 > 1: \text{divergent}]$$

$$(g) \sum_{n=1}^{\infty} \frac{n^p}{n!}. \quad [\text{Ans. } L = 0 < 1: \text{convergent for every real } p]$$

$$(h) \sum_{n=1}^{\infty} \frac{n!}{4^{n-3}}. \quad [\text{Ans. } L = \infty: \text{divergent}]$$

$$(i) \sum_{n=1}^{\infty} \frac{8n-4}{9n+9}.$$

Ans. $L = 1$, so the ratio test fails; since $a_n \rightarrow \frac{8}{9} \neq 0$, the series diverges by the n th term test.

3. Examine the convergence of the series $\sum_{n=1}^{\infty} \left(\frac{nx}{n+1} \right)^n$, where x is real. $\blacktriangleright$ Problem 3.7
 [Ans. convergent for $|x| < 1$, divergent for $|x| \geq 1$]

4. Discuss the convergence of the series $1 + \frac{2!}{2^2} + \frac{3!}{3^3} + \frac{4!}{4^4} + \dots$.
 [Ans. $a_n = \frac{n!}{n^n}$, $L = \frac{1}{e} < 1$: convergent]

5. Discuss the convergence of the series $\frac{x}{1+x} + \frac{x^2}{1+x^2} + \frac{x^3}{1+x^3} + \dots + \frac{x^n}{1+x^n} + \dots$, where $x > 0$.
 [Ans. convergent for $0 < x < 1$, divergent for $x \geq 1$]

Taylor and Maclaurin series

1. Obtain the Maclaurin series expansion, as an infinite series, of each of the following functions. $\blacktriangleright$ Problem 4.1

$$(a) \sin x. \quad [\text{Ans. } \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}]$$

$$(b) \cos x. \quad [\text{Ans. } \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}]$$

$$(c) e^x. \quad [\text{Ans. } \sum_{n=0}^{\infty} \frac{x^n}{n!}]$$

$$(d) \log(1+x). \quad [\text{Ans. } \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n}]$$

$$(e) (1+x)^k.$$

$$\text{Ans. } 1 + kx + \frac{k(k-1)}{2!}x^2 + \frac{k(k-1)(k-2)}{3!}x^3 + \frac{k(k-1)(k-2)(k-3)}{4!}x^4 + \dots$$

2. Obtain the Taylor series expansion of $\log_e x$ about $x = 1$ up to the term containing $(x-1)^4$, and hence find the approximate value of $\log_e(1.1)$. $\blacktriangleright$ Problem 4.2

$$\text{Ans. } (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \frac{(x-1)^4}{4}; \log(1.1) \approx 0.0953$$

3. Obtain the Taylor series expansion of $\log(\cos x)$ about $x = \frac{\pi}{3}$ up to the fourth-degree term.

Ans. $-\log 2 - \sqrt{3} \left(x - \frac{\pi}{3}\right) - 2 \left(x - \frac{\pi}{3}\right)^2 - \frac{4}{\sqrt{3}} \left(x - \frac{\pi}{3}\right)^3 - \frac{10}{3} \left(x - \frac{\pi}{3}\right)^4$

4. Obtain the Taylor series expansion of $f(x) = \sin x$ about $x = \frac{\pi}{6}$ up to the term containing $\left(x - \frac{\pi}{6}\right)^5$. ► Problem 4.3

Ans. $\frac{1}{2} + \frac{\sqrt{3}}{2} \left(x - \frac{\pi}{6}\right) - \frac{1}{4} \left(x - \frac{\pi}{6}\right)^2 - \frac{\sqrt{3}}{12} \left(x - \frac{\pi}{6}\right)^3 + \frac{1}{48} \left(x - \frac{\pi}{6}\right)^4 + \frac{\sqrt{3}}{240} \left(x - \frac{\pi}{6}\right)^5$

5. Expand $\tan^{-1} x$ in powers of $(x - 1)$ up to the term containing $(x - 1)^4$.

Ans. $\frac{\pi}{4} + \frac{1}{2} \left[(x - 1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{6} \right]$ (the $(x - 1)^4$ term is 0)

6. Expand $\log(\sec x)$ up to the term containing x^6 using Maclaurin's series.

[**Ans.** $\frac{x^2}{2} + \frac{x^4}{12} + \frac{x^6}{45}$]

7. Expand $e^{\sin x}$ by Maclaurin's series up to the term containing x^4 . [**Ans.** $1 + x + \frac{x^2}{2} - \frac{x^4}{8}$]

8. Obtain the Maclaurin expansion of a^x up to the third-degree term.

Ans. $1 + (\log a)x + \frac{(\log a)^2}{2}x^2 + \frac{(\log a)^3}{6}x^3$

9. Obtain $\log(1 + \cos x)$ by Maclaurin's series up to the term containing x^4 .

[**Ans.** $\log 2 - \frac{x^2}{4} - \frac{x^4}{96}$]

10. Expand $\tan x$ about $x = \frac{\pi}{4}$ up to the third-degree term, and hence find $\tan 46^\circ$.

Ans. $1 + 2 \left(x - \frac{\pi}{4}\right) + 2 \left(x - \frac{\pi}{4}\right)^2 + \frac{8}{3} \left(x - \frac{\pi}{4}\right)^3$; $\tan 46^\circ \approx 1.0355$

11. Evaluate $\int e^{-x^2} dx$ as an infinite series, using the Maclaurin series of e^{-x^2} ; write the terms of e^{-x^2} up to x^4 explicitly. ► Problem 4.4

[**Ans.** $x - \frac{x^3}{3} + \frac{x^5}{10} - \dots + C$]

1 Rolle's Theorem

Theorem 1.1: Rolle's theorem

Let f be a function that satisfies the following three conditions:

- (i) f is continuous on the closed interval $[a, b]$;
- (ii) f is differentiable on the open interval (a, b) ;
- (iii) $f(a) = f(b)$.

Then there is a number $c \in (a, b)$ such that $f'(c) = 0$.

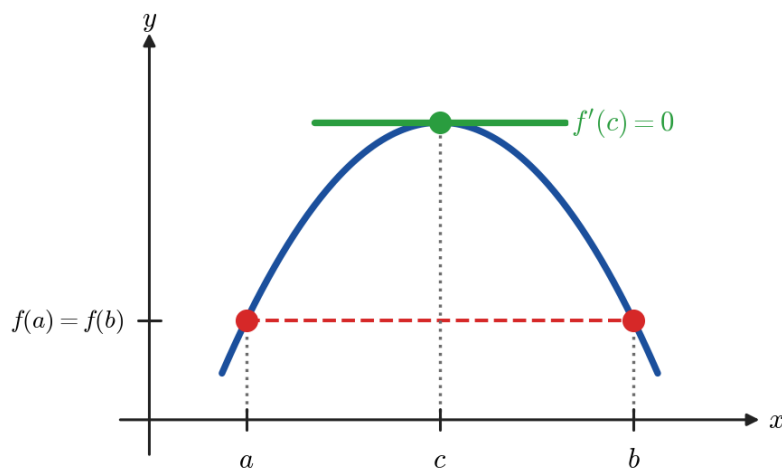


Figure 1.1: When the curve starts and ends at the same height, it must turn somewhere in between, and there the tangent is horizontal.

Note:

How to verify Rolle's theorem.

1. Write down the given data: $f(x)$, a and b .
2. Check conditions (i) and (ii), and evaluate $f(a)$ and $f(b)$ by substitution to check (iii).
3. Differentiate to get $f'(x)$, and write $f'(c)$.
4. Solve $f'(c) = 0$ and keep only the solutions that lie strictly inside (a, b) .

If a condition fails, the theorem does not apply — and then a point with $f'(c) = 0$ may or may not exist.

Note:

Sums, products, quotients (with non-zero denominator) and composites of polynomials, exponential, logarithmic and trigonometric functions are continuous and differentiable wherever they are defined. This settles conditions (i) and (ii) for most functions met in practice.

1.1 Worked problems

Problem 1.1. [Tutorial 1, Q1(a)] Verify Rolle's theorem for $f(x) = x^2(1-x)^2$ on $[0, 1]$.

Solution: Given: $f(x) = x^2(1-x)^2$, $a = 0$, $b = 1$.

Checking the conditions.

- (i) f is a polynomial, so it is continuous on $[0, 1]$.
- (ii) Being a polynomial, f is differentiable on $(0, 1)$.
- (iii) Substituting the end points,

$$f(a) = f(0) = 0^2(1-0)^2 = 0 \cdot 1 = 0, \quad f(b) = f(1) = 1^2(1-1)^2 = 1 \cdot 0 = 0,$$

$$\text{so } f(a) = f(b).$$

All three conditions hold.

Finding c . Multiplying out first makes the differentiation easy:

$$f(x) = x^2(1-2x+x^2) = x^2 - 2x^3 + x^4.$$

Hence

$$f'(x) = 2x - 6x^2 + 4x^3 \quad \text{and} \quad f'(c) = 2c - 6c^2 + 4c^3.$$

By Rolle's theorem, $f'(c) = 0$:

$$2c - 6c^2 + 4c^3 = 0 \implies 2c(2c^2 - 3c + 1) = 0 \implies 2c(2c-1)(c-1) = 0,$$

so $c = 0$, $c = \frac{1}{2}$ or $c = 1$. Only $c = \frac{1}{2}$ lies in the open interval $(0, 1)$. As a check,

$$f'\left(\frac{1}{2}\right) = 2\left(\frac{1}{2}\right) - 6\left(\frac{1}{4}\right) + 4\left(\frac{1}{8}\right) = 1 - \frac{3}{2} + \frac{1}{2} = 0.$$

Hence Rolle's theorem is verified, with $c = \frac{1}{2}$.

Problem 1.2. [Tutorial 1, Q1(d)] Verify Rolle's theorem for $f(x) = e^x \sin x$ on $[0, \pi]$.

Solution: Given: $f(x) = e^x \sin x$, $a = 0$, $b = \pi$.

Checking the conditions.

- (i) e^x and $\sin x$ are continuous everywhere, so their product is continuous on $[0, \pi]$.
- (ii) Both are differentiable everywhere, so f is differentiable on $(0, \pi)$.
- (iii) Substituting the end points,

$$f(a) = f(0) = e^0 \sin 0 = 1 \cdot 0 = 0, \quad f(b) = f(\pi) = e^\pi \sin \pi = e^\pi \cdot 0 = 0,$$

$$\text{so } f(a) = f(b).$$

All three conditions hold.

Finding c . By the product rule,

$$f'(x) = e^x \sin x + e^x \cos x = e^x(\sin x + \cos x), \quad f'(c) = e^c(\sin c + \cos c).$$

By Rolle's theorem, $f'(c) = 0$. Since $e^c > 0$,

$$\sin c + \cos c = 0 \implies \sin c = -\cos c \implies \tan c = -1.$$

In $(0, \pi)$ the tangent is negative only in the second quadrant, so $c = \pi - \frac{\pi}{4} = \frac{3\pi}{4} \in (0, \pi)$. As a check, $f'\left(\frac{3\pi}{4}\right) = e^{3\pi/4} \left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}\right) = 0$. Hence Rolle's theorem is verified, with $c = \frac{3\pi}{4}$.

Problem 1.3. [Tutorial 1, Q1(e)] Verify Rolle's theorem for $f(x) = \log\left(\frac{x^2 + 9}{10x}\right)$ on $[1, 9]$.

Solution: Given: $f(x) = \log\left(\frac{x^2 + 9}{10x}\right)$, $a = 1$, $b = 9$.

Checking the conditions.

- (i) For $x \in [1, 9]$ the quantity $\frac{x^2 + 9}{10x}$ is continuous and positive, so its logarithm is defined and continuous on $[1, 9]$.
- (ii) For the same reason f is differentiable on $(1, 9)$.
- (iii) Substituting the end points,

$$f(a) = f(1) = \log\left(\frac{1^2 + 9}{10(1)}\right) = \log \frac{10}{10} = \log 1 = 0,$$

$$f(b) = f(9) = \log\left(\frac{9^2 + 9}{10(9)}\right) = \log \frac{90}{90} = \log 1 = 0,$$

$$\text{so } f(a) = f(b).$$

All three conditions hold.

Finding c . Before differentiating, simplify using $\log \frac{A}{B} = \log A - \log B$ and $\log(10x) = \log 10 + \log x$:

$$f(x) = \log(x^2 + 9) - \log 10 - \log x.$$

Differentiating term by term ($\log 10$ is a constant),

$$f'(x) = \frac{2x}{x^2 + 9} - 0 - \frac{1}{x}, \quad f'(c) = \frac{2c}{c^2 + 9} - \frac{1}{c} = \frac{2c^2 - (c^2 + 9)}{c(c^2 + 9)} = \frac{c^2 - 9}{c(c^2 + 9)}.$$

By Rolle's theorem, $f'(c) = 0$, so the numerator must vanish:

$$c^2 - 9 = 0 \implies c = \pm 3.$$

Only $c = 3$ lies in $(1, 9)$. Hence Rolle's theorem is verified, with $c = 3$.

Problem 1.4. [Tutorial 1, Q1(g)] Verify Rolle's theorem for $f(x) = \sqrt[3]{(x-2)^2}$ on $[1, 3]$.

Solution: Given: $f(x) = \sqrt[3]{(x-2)^2} = (x-2)^{2/3}$, $a = 1$, $b = 3$.

Checking the conditions.

- (i) f is the cube root of a polynomial, so it is continuous on $[1, 3]$.
- (ii) Differentiating,

$$f'(x) = \frac{2}{3}(x-2)^{-1/3} = \frac{2}{3\sqrt[3]{x-2}},$$

which is not defined at $x = 2$. Since $2 \in (1, 3)$, f is *not* differentiable on $(1, 3)$.

(iii) Substituting the end points,

$$f(a) = f(1) = \sqrt[3]{(1-2)^2} = \sqrt[3]{(-1)^2} = \sqrt[3]{1} = 1,$$

$$f(b) = f(3) = \sqrt[3]{(3-2)^2} = \sqrt[3]{1^2} = \sqrt[3]{1} = 1,$$

so $f(a) = f(b)$.

Condition (ii) fails, so Rolle's theorem does not apply. The conclusion fails as well: wherever it exists,

$$f'(c) = \frac{2}{3\sqrt[3]{c-2}},$$

and $f'(c) = 0$ would need the numerator 2 to be zero, which is impossible. There is no $c \in (1, 3)$ with $f'(c) = 0$.

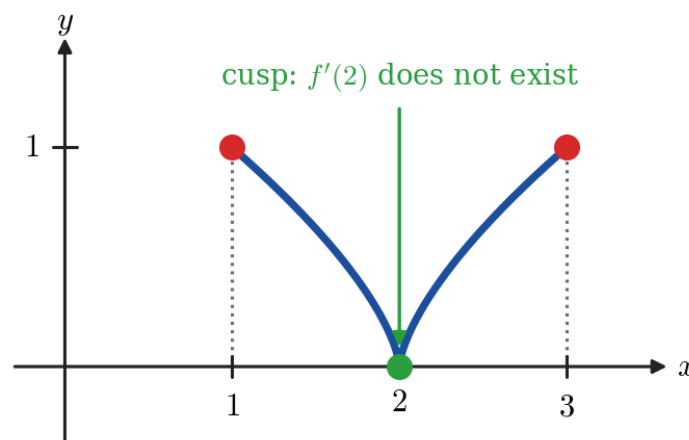


Figure 1.2: $f(x) = \sqrt[3]{(x-2)^2}$ on $[1, 3]$. The ends are at the same height, but the curve turns at a sharp point rather than smoothly, so there is no horizontal tangent.

Note:

This is why the differentiability condition cannot be dropped: a single point where the derivative fails to exist is enough to destroy the conclusion.

Problem 1.5. [Tutorial 1, Q6] A ball is thrown vertically upward from the ground with an initial speed of 29.4 m/s. Its height above the ground after t seconds is

$$h(t) = 29.4t - 4.9t^2 \quad \text{metres.}$$

Show that at some instant during the flight the ball is momentarily at rest, and find that instant and the maximum height attained.

Solution: Given: $h(t) = 29.4t - 4.9t^2$. The interval is the duration of the flight, which starts and ends when the ball is on the ground, $h(t) = 0$:

$$29.4t - 4.9t^2 = 4.9t(6 - t) = 0 \implies t = 0 \quad \text{or} \quad t = 6.$$

So $a = 0$ and $b = 6$.

Checking the conditions.

(i) h is a polynomial, so it is continuous on $[0, 6]$.

(ii) It is differentiable on $(0, 6)$.

(iii) Substituting the end points,

$$h(a) = h(0) = 29.4(0) - 4.9(0)^2 = 0,$$

$$h(b) = h(6) = 29.4(6) - 4.9(6)^2 = 176.4 - 4.9(36) = 176.4 - 176.4 = 0,$$

$$\text{so } h(a) = h(b).$$

By Rolle's theorem there is an instant $c \in (0, 6)$ with $h'(c) = 0$. Since $h'(t)$ is the velocity of the ball, the ball is momentarily at rest at $t = c$.

Finding c .

$$h'(t) = 29.4 - 9.8t, \quad h'(c) = 29.4 - 9.8c.$$

Setting $h'(c) = 0$ gives $c = \frac{29.4}{9.8} = 3$ s, which lies in $(0, 6)$. Since $h''(t) = -9.8 < 0$, this instant gives the maximum height:

$$h(3) = 29.4(3) - 4.9(3)^2 = 88.2 - 44.1 = 44.1 \text{ m.}$$

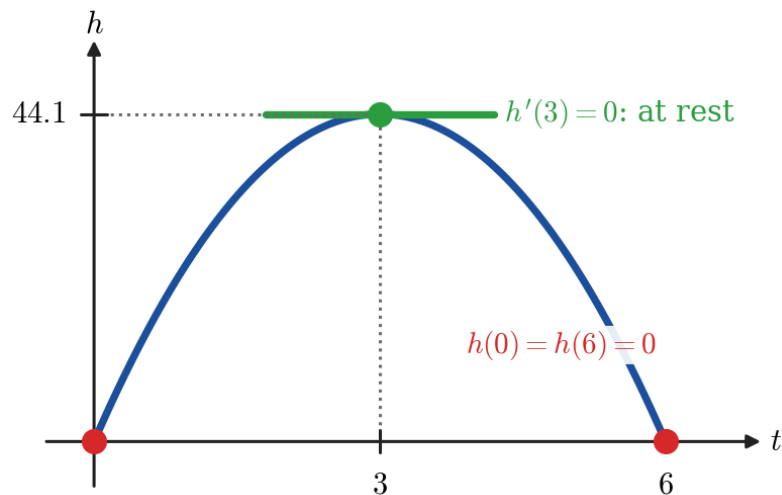


Figure 1.3: Height of the ball. It leaves and returns to the ground, so by Rolle's theorem its velocity $h'(t)$ is zero at some instant: the top of the flight, $t = 3$ s.

Problem 1.6. [Tutorial 1, Q8] In an automated chemical batch reactor, an exothermic reaction is monitored over an 8-minute cycle, $t \in [0, 8]$. The temperature $T(t)$, in $^{\circ}\text{C}$, is given by

$$T(t) = 50 + 12t - 1.5t^2.$$

- Verify that T satisfies all the conditions of Rolle's theorem on $[0, 8]$.
- Determine the critical time $c \in (0, 8)$ at which the instantaneous rate of temperature change vanishes, and find the maximum temperature.

Solution: Given: $T(t) = 50 + 12t - 1.5t^2$, $a = 0$, $b = 8$.

- T is a polynomial, so it is continuous on $[0, 8]$ and differentiable on $(0, 8)$. Substituting the end points,

$$T(a) = T(0) = 50 + 12(0) - 1.5(0)^2 = 50,$$

$$T(b) = T(8) = 50 + 12(8) - 1.5(8)^2 = 50 + 96 - 96 = 50,$$

so $T(a) = T(b) = 50^{\circ}\text{C}$. All three conditions of Rolle's theorem hold.

(b) The instantaneous rate of change of temperature is

$$T'(t) = 12 - 3t, \quad T'(c) = 12 - 3c.$$

By Rolle's theorem, $T'(c) = 0$, so $12 - 3c = 0$ and $c = 4$ minutes, which lies in $(0, 8)$. Since $T''(t) = -3 < 0$, the temperature is greatest at this instant:

$$T(4) = 50 + 12(4) - 1.5(4)^2 = 50 + 48 - 24 = 74^\circ\text{C}.$$

2 Lagrange's Mean Value Theorem

Theorem 2.1: Lagrange's mean value theorem

Let f be a function that satisfies the following conditions:

- (i) f is continuous on the closed interval $[a, b]$;
- (ii) f is differentiable on the open interval (a, b) .

Then there is a number $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}. \quad (2.1)$$

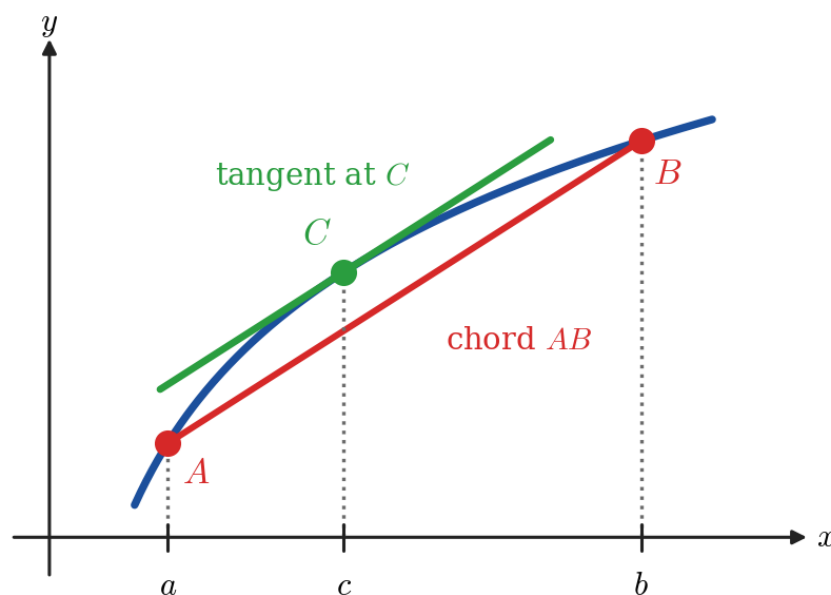


Figure 2.1: At some point C between A and B the tangent is parallel to the chord AB : its slope $f'(c)$ equals the slope $\frac{f(b)-f(a)}{b-a}$ of the chord.

Note:

How to verify Lagrange's theorem.

1. Write down the given data: $f(x)$, a and b .
2. Check conditions (i) and (ii).
3. Evaluate $f(a)$ and $f(b)$ by substitution, and compute the slope $\frac{f(b)-f(a)}{b-a}$ of the chord.
4. Differentiate to get $f'(x)$, write $f'(c)$, and solve equation (2.1), keeping only solutions in (a, b) .

Note:

Multiplying (2.1) by $b - a$ gives $f(b) - f(a) = (b - a) f'(c)$. This turns information about the derivative into information about the function: for instance, if $f'(x) \leq M$ throughout (a, b) , then $f(b) - f(a) \leq M(b - a)$. This is how the theorem proves inequalities and bounds.

2.1 Worked problems

Problem 2.1. [Tutorial 1, Q3(a)] Verify Lagrange's mean value theorem for $f(x) = \cos^2 x$ on $\left[0, \frac{\pi}{2}\right]$.

Solution: Given: $f(x) = \cos^2 x$, $a = 0$, $b = \frac{\pi}{2}$.

Checking the conditions. $\cos^2 x$ is continuous and differentiable everywhere, so it is continuous on $\left[0, \frac{\pi}{2}\right]$ and differentiable on $\left(0, \frac{\pi}{2}\right)$.

Slope of the chord. Substituting the end points,

$$f(a) = f(0) = \cos^2 0 = 1^2 = 1, \quad f(b) = f\left(\frac{\pi}{2}\right) = \cos^2 \frac{\pi}{2} = 0^2 = 0,$$

so

$$\frac{f(b) - f(a)}{b - a} = \frac{0 - 1}{\frac{\pi}{2} - 0} = -\frac{2}{\pi}.$$

Finding c .

$$f'(x) = 2 \cos x (-\sin x) = -\sin 2x, \quad f'(c) = -\sin 2c.$$

By (2.1),

$$-\sin 2c = -\frac{2}{\pi} \implies \sin 2c = \frac{2}{\pi}.$$

For $c \in \left(0, \frac{\pi}{2}\right)$ we have $2c \in (0, \pi)$, where the sine takes each value in $(0, 1)$ twice:

$$2c = \sin^{-1}\left(\frac{2}{\pi}\right) \quad \text{or} \quad 2c = \pi - \sin^{-1}\left(\frac{2}{\pi}\right).$$

Hence

$$c = \frac{1}{2} \sin^{-1}\left(\frac{2}{\pi}\right) \approx 0.3451 \quad \text{or} \quad c = \frac{1}{2} \left[\pi - \sin^{-1}\left(\frac{2}{\pi}\right) \right] \approx 1.2257.$$

Both values lie in $\left(0, \frac{\pi}{2}\right) \approx (0, 1.5708)$, so the theorem is verified — here with two admissible points.

Problem 2.2. [Tutorial 1, Q3(b)] Verify Lagrange's mean value theorem for $f(x) = \sin^{-1} x$ on $[0, 1]$.

Solution: Given: $f(x) = \sin^{-1} x$, $a = 0$, $b = 1$.

Checking the conditions. $\sin^{-1} x$ is continuous on $[-1, 1]$, hence on $[0, 1]$. Its derivative $\frac{1}{\sqrt{1-x^2}}$ exists for $-1 < x < 1$, so f is differentiable on $(0, 1)$. (It is not differentiable at $x = 1$, but the theorem asks only for differentiability on the *open* interval.)

Slope of the chord. Substituting the end points,

$$f(a) = f(0) = \sin^{-1} 0 = 0, \quad f(b) = f(1) = \sin^{-1} 1 = \frac{\pi}{2},$$

so

$$\frac{f(b) - f(a)}{b - a} = \frac{\frac{\pi}{2} - 0}{1 - 0} = \frac{\pi}{2}.$$

Finding c .

$$f'(x) = \frac{1}{\sqrt{1-x^2}}, \quad f'(c) = \frac{1}{\sqrt{1-c^2}}.$$

By (2.1),

$$\frac{1}{\sqrt{1-c^2}} = \frac{\pi}{2} \implies \sqrt{1-c^2} = \frac{2}{\pi} \implies 1-c^2 = \frac{4}{\pi^2} \implies c = \sqrt{1 - \frac{4}{\pi^2}} = \frac{\sqrt{\pi^2 - 4}}{\pi} \approx 0.7712,$$

taking the positive root since $c \in (0, 1)$. The theorem is verified.

Problem 2.3. [Tutorial 1, Q3(c)] Verify Lagrange's mean value theorem for $f(x) = 2x^3 - 3x + 1$ on $[-2, 2]$.

Solution: Given: $f(x) = 2x^3 - 3x + 1$, $a = -2$, $b = 2$.

Checking the conditions. f is a polynomial, so it is continuous on $[-2, 2]$ and differentiable on $(-2, 2)$.

Slope of the chord. Substituting the end points,

$$f(a) = f(-2) = 2(-2)^3 - 3(-2) + 1 = -16 + 6 + 1 = -9,$$

$$f(b) = f(2) = 2(2)^3 - 3(2) + 1 = 16 - 6 + 1 = 11,$$

so

$$\frac{f(b) - f(a)}{b - a} = \frac{11 - (-9)}{2 - (-2)} = \frac{20}{4} = 5.$$

Finding c .

$$f'(x) = 6x^2 - 3, \quad f'(c) = 6c^2 - 3.$$

By (2.1),

$$6c^2 - 3 = 5 \implies 6c^2 = 8 \implies c^2 = \frac{4}{3} \implies c = \pm \frac{2}{\sqrt{3}} \approx \pm 1.1547.$$

Both values lie in $(-2, 2)$, so the theorem is verified, with two admissible points.

Problem 2.4. [Tutorial 1, Q4] Apply Lagrange's mean value theorem to $f(x) = \log x$ to prove that

$$0 < \frac{1}{\log(1+x)} - \frac{1}{x} < 1 \quad \text{for } x > 0.$$

Solution: Given: $f(t) = \log t$ on $[1, 1+x]$, where $x > 0$; so $a = 1$ and $b = 1+x$. (The variable of f is written t , since x is already in use.)

Checking the conditions. $\log t$ is continuous and differentiable for $t > 0$, so it is continuous on $[1, 1+x]$ and differentiable on $(1, 1+x)$.

Applying the theorem. Substituting the end points,

$$f(a) = f(1) = \log 1 = 0, \quad f(b) = f(1+x) = \log(1+x),$$

and

$$f'(t) = \frac{1}{t}, \quad f'(c) = \frac{1}{c}.$$

By (2.1), there is a $c \in (1, 1+x)$ such that

$$\frac{1}{c} = \frac{\log(1+x) - 0}{(1+x) - 1} = \frac{\log(1+x)}{x} \implies c = \frac{x}{\log(1+x)}.$$

(Here $\log(1+x) > 0$ because $1+x > 1$.)

Using $c \in (1, 1+x)$. Since $1 < c < 1+x$,

$$1 < \frac{x}{\log(1+x)} < 1+x.$$

Dividing throughout by $x > 0$, which keeps the inequalities in the same direction,

$$\frac{1}{x} < \frac{1}{\log(1+x)} < \frac{1}{x} + 1.$$

Finally, subtracting $\frac{1}{x}$ throughout,

$$0 < \frac{1}{\log(1+x)} - \frac{1}{x} < 1,$$

which is the required inequality.

Problem 2.5. [Tutorial 1, Q12] Suppose that f is continuous on $[-7, 0]$ and differentiable on $(-7, 0)$, with $f(-7) = -3$ and $f'(x) \leq 2$ for all $x \in (-7, 0)$. Determine the largest possible value of $f(0)$.

Solution: Given: $a = -7$, $b = 0$, $f(a) = f(-7) = -3$, and $f'(x) \leq 2$ on $(-7, 0)$. The conditions of Lagrange's theorem are part of the data.

Applying the theorem. By (2.1), there is a $c \in (-7, 0)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a} = \frac{f(0) - f(-7)}{0 - (-7)} = \frac{f(0) - (-3)}{7} = \frac{f(0) + 3}{7}.$$

Since $f'(c) \leq 2$,

$$\frac{f(0) + 3}{7} \leq 2 \implies f(0) + 3 \leq 14 \implies f(0) \leq 11.$$

The bound is attained. The function $f(x) = 2x + 11$ satisfies all the conditions:

$$f(-7) = 2(-7) + 11 = -14 + 11 = -3, \quad f'(x) = 2 \leq 2,$$

and it gives $f(0) = 2(0) + 11 = 11$. Hence the largest possible value of $f(0)$ is 11.

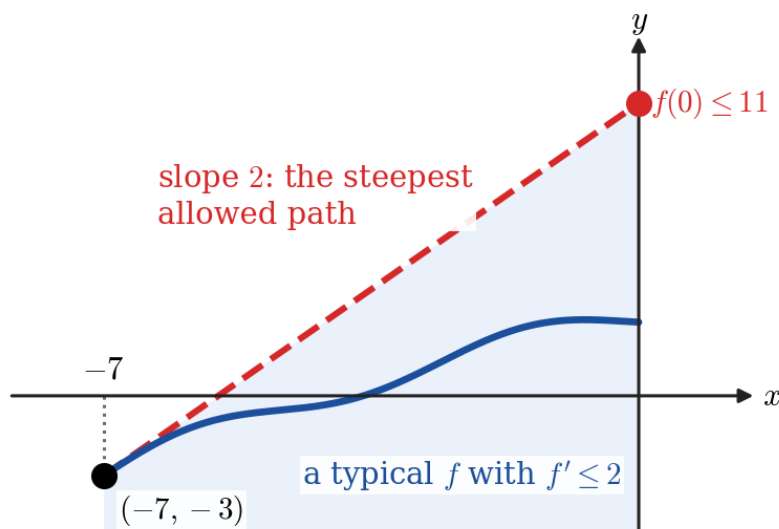


Figure 2.2: Starting from $(-7, -3)$ with slope never exceeding 2, no graph can rise above the line of slope 2; so $f(0) \leq 11$, with equality only for $f(x) = 2x + 11$.

3 Series: the Root and Ratio Tests

Definition 3.1: Convergence of a series

Given a sequence $\{a_n\}$, the series $\sum_{n=1}^{\infty} a_n$ has partial sums $S_n = a_1 + a_2 + \cdots + a_n$. The series *converges* if $\lim_{n \rightarrow \infty} S_n$ exists as a finite number, and *diverges* otherwise.

Theorem 3.2: The n th term test

If $\lim_{n \rightarrow \infty} a_n \neq 0$ (or the limit does not exist), then $\sum a_n$ diverges.

Note:

The converse is false: $a_n \rightarrow 0$ does *not* by itself make $\sum a_n$ converge. The n th term test can only prove divergence.

Theorem 3.3: The root test

Let $L = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \lim_{n \rightarrow \infty} |a_n|^{1/n}$.

- (i) If $L < 1$, the series $\sum a_n$ converges (absolutely).
- (ii) If $L > 1$ (including $L = \infty$), the series diverges.
- (iii) If $L = 1$, the test gives no information.

Theorem 3.4: The ratio test

Let $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$.

- (i) If $L < 1$, the series $\sum a_n$ converges (absolutely).
- (ii) If $L > 1$ (including $L = \infty$), the series diverges.
- (iii) If $L = 1$, the test gives no information.

Note:

Which test? Use the root test when the general term is an n th power, such as $\left(\frac{2n+3}{3n+2}\right)^n$. Use the ratio test when it contains factorials, or powers such as 2^n and 3^n multiplied by polynomials or logarithms. In both tests, first simplify $|a_n|^{1/n}$ or $\left|\frac{a_{n+1}}{a_n}\right|$ as an expression in n , and only then take the limit.

3.1 Standard limits of sequences

- (i) If $p > 0$, then $\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0$.
- (ii) If $p > 0$, then $\lim_{n \rightarrow \infty} p^{1/n} = 1$.
- (iii) $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$, and more generally $\lim_{n \rightarrow \infty} \left(1 + \frac{k}{n}\right)^n = e^k$ for any constant k .

(iv) $\lim_{n \rightarrow \infty} n^{1/n} = 1.$

(v) If $|x| < 1$, then $\lim_{n \rightarrow \infty} x^n = 0$; in particular $\frac{1}{r^n} = \left(\frac{1}{r}\right)^n \rightarrow 0$ when $r > 1$.

(vi) If $\alpha, \beta > 0$, then $\lim_{n \rightarrow \infty} \frac{n^\alpha}{(1 + \beta)^n} = 0.$

(vii) $\lim_{n \rightarrow \infty} (n!)^{1/n} = \infty$, where $n! = n(n-1)(n-2) \cdots 3 \cdot 2 \cdot 1$.

Note:

In (vi) the numerator n^α is a power function and the denominator $(1 + \beta)^n$ is an exponential function. The exponential grows faster, so the ratio tends to 0.

3.2 Worked problems on the root test

Problem 3.1. [Tutorial 2, Series Q1(a)] Test the convergence of $\sum_{n=1}^{\infty} \left(\frac{2n+3}{3n+2}\right)^n$ by the root test.

Solution: Given: $a_n = \left(\frac{2n+3}{3n+2}\right)^n$, which is positive, so $|a_n| = a_n$.

Computing $|a_n|^{1/n}$.

$$|a_n|^{1/n} = \left[\left(\frac{2n+3}{3n+2}\right)^n\right]^{1/n} = \frac{2n+3}{3n+2}.$$

Taking the limit. Dividing the numerator and denominator by n and using standard limit (i),

$$L = \lim_{n \rightarrow \infty} \frac{2n+3}{3n+2} = \lim_{n \rightarrow \infty} \frac{2 + \frac{3}{n}}{3 + \frac{2}{n}} = \frac{2+0}{3+0} = \frac{2}{3}.$$

Since $L = \frac{2}{3} < 1$, the series is convergent by the root test.

Problem 3.2. [Tutorial 2, Series Q1(b)] Test the convergence of $\sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^{-n^2}$ by the root test.

Solution: Given: $a_n = \left(1 + \frac{1}{n}\right)^{-n^2}$, which is positive, so $|a_n| = a_n$.

Computing $|a_n|^{1/n}$. Raising to the power $\frac{1}{n}$ divides the exponent $-n^2$ by n :

$$|a_n|^{1/n} = \left[\left(1 + \frac{1}{n}\right)^{-n^2}\right]^{1/n} = \left(1 + \frac{1}{n}\right)^{-n} = \frac{1}{\left(1 + \frac{1}{n}\right)^n}.$$

Taking the limit. By standard limit (iii),

$$L = \lim_{n \rightarrow \infty} \frac{1}{\left(1 + \frac{1}{n}\right)^n} = \frac{1}{e}.$$

Since $L = \frac{1}{e} \approx 0.368 < 1$, the series is convergent by the root test.

Problem 3.3. [Tutorial 2, Series Q1(h)] Test the convergence of $\sum_{n=1}^{\infty} \left(1 + \frac{2}{n}\right)^{n^2}$ by the root test.

Solution: Given: $a_n = \left(1 + \frac{2}{n}\right)^{n^2}$, which is positive, so $|a_n| = a_n$.

Computing $|a_n|^{1/n}$.

$$|a_n|^{1/n} = \left[\left(1 + \frac{2}{n}\right)^{n^2}\right]^{1/n} = \left(1 + \frac{2}{n}\right)^n.$$

Taking the limit. By standard limit (iii) with $k = 2$,

$$L = \lim_{n \rightarrow \infty} \left(1 + \frac{2}{n}\right)^n = e^2.$$

Since $L = e^2 \approx 7.39 > 1$, the series is divergent by the root test.

3.3 Worked problems on the ratio test

Problem 3.4. [Tutorial 2, Series Q2(a)] Test the convergence of $\sum_{n=1}^{\infty} \frac{2^{n-1}}{3^n + 1}$ by the ratio test.

Solution: Given: $a_n = \frac{2^{n-1}}{3^n + 1}$. Replacing n by $n + 1$,

$$a_{n+1} = \frac{2^{(n+1)-1}}{3^{n+1} + 1} = \frac{2^n}{3^{n+1} + 1}.$$

Computing $\left|\frac{a_{n+1}}{a_n}\right|$.

$$\left|\frac{a_{n+1}}{a_n}\right| = \frac{2^n}{3^{n+1} + 1} \cdot \frac{3^n + 1}{2^{n-1}} = \frac{2^n}{2^{n-1}} \cdot \frac{3^n + 1}{3^{n+1} + 1} = \frac{2(3^n + 1)}{3^{n+1} + 1}.$$

Taking the limit. Dividing the numerator and denominator by 3^n , and using standard limit (v),

$$L = \lim_{n \rightarrow \infty} \frac{2\left(1 + \frac{1}{3^n}\right)}{3 + \frac{1}{3^n}} = \frac{2(1 + 0)}{3 + 0} = \frac{2}{3}.$$

Since $L = \frac{2}{3} < 1$, the series is convergent by the ratio test.

Problem 3.5. [Tutorial 2, Series Q2(d)] Test the convergence of $\sum_{n=0}^{\infty} \frac{n^3 + a}{2^n + a}$, where $a > 0$ is a constant, by the ratio test.

Solution: Given: $a_n = \frac{n^3 + a}{2^n + a}$, which is positive. Replacing n by $n + 1$,

$$a_{n+1} = \frac{(n+1)^3 + a}{2^{n+1} + a}.$$

Computing $\left|\frac{a_{n+1}}{a_n}\right|$.

$$\left|\frac{a_{n+1}}{a_n}\right| = \frac{(n+1)^3 + a}{2^{n+1} + a} \cdot \frac{2^n + a}{n^3 + a} = \frac{(n+1)^3 + a}{n^3 + a} \cdot \frac{2^n + a}{2^{n+1} + a}.$$

Dividing by n^3 in the first factor and by 2^n in the second,

$$\left|\frac{a_{n+1}}{a_n}\right| = \frac{\left(1 + \frac{1}{n}\right)^3 + \frac{a}{n^3}}{1 + \frac{a}{n^3}} \cdot \frac{1 + \frac{a}{2^n}}{2 + \frac{a}{2^n}}.$$

Taking the limit. By standard limits (i) and (v), $\frac{1}{n} \rightarrow 0$, $\frac{a}{n^3} \rightarrow 0$ and $\frac{a}{2^n} \rightarrow 0$, so

$$L = \frac{(1+0)^3 + 0}{1+0} \cdot \frac{1+0}{2+0} = 1 \cdot \frac{1}{2} = \frac{1}{2}.$$

Since $L = \frac{1}{2} < 1$, the series is convergent by the ratio test.

Note:

The polynomial factor contributes only a limit of 1: in a ratio test, the exponential 2^n decides the outcome against any power of n .

Problem 3.6. [Tutorial 2, Series Q2(f)] Test the convergence of $\sum_{n=2}^{\infty} \frac{3^{n+2}}{\log n}$ by the ratio test.

Solution: Given: $a_n = \frac{3^{n+2}}{\log n}$, which is positive for $n \geq 2$. Replacing n by $n+1$,

$$a_{n+1} = \frac{3^{(n+1)+2}}{\log(n+1)} = \frac{3^{n+3}}{\log(n+1)}.$$

Computing $\left| \frac{a_{n+1}}{a_n} \right|$.

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{3^{n+3}}{\log(n+1)} \cdot \frac{\log n}{3^{n+2}} = 3 \frac{\log n}{\log(n+1)}.$$

Since $\log(n+1) = \log \left[n \left(1 + \frac{1}{n} \right) \right] = \log n + \log \left(1 + \frac{1}{n} \right)$, dividing the numerator and denominator by $\log n$ gives

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{3}{1 + \frac{\log \left(1 + \frac{1}{n} \right)}{\log n}}.$$

Taking the limit. As $n \rightarrow \infty$, $\log \left(1 + \frac{1}{n} \right) \rightarrow \log 1 = 0$ while $\log n \rightarrow \infty$, so

$$L = \frac{3}{1+0} = 3.$$

Since $L = 3 > 1$, the series is divergent by the ratio test. (This agrees with the n th term test: $a_n \rightarrow \infty$.)

3.4 A series depending on a variable

When the general term contains a variable x , the test returns a limit L that depends on x . The conditions $L < 1$ and $L > 1$ then become conditions on x , and the values of x with $L = 1$ must be examined separately.

Problem 3.7. [Tutorial 2, Series Q3] Examine the convergence of the series $\sum_{n=1}^{\infty} \left(\frac{nx}{n+1} \right)^n$, where x is real.

Solution: Given: $a_n = \left(\frac{nx}{n+1} \right)^n$, so $|a_n| = \left(\frac{n|x|}{n+1} \right)^n$.

Computing $|a_n|^{1/n}$.

$$|a_n|^{1/n} = \left[\left(\frac{n|x|}{n+1} \right)^n \right]^{1/n} = \frac{n}{n+1} |x| = \frac{|x|}{1 + \frac{1}{n}}.$$

Taking the limit. By standard limit (i),

$$L = \lim_{n \rightarrow \infty} \frac{|x|}{1 + \frac{1}{n}} = \frac{|x|}{1 + 0} = |x|.$$

- If $|x| < 1$, then $L < 1$ and the series is convergent.
- If $|x| > 1$, then $L > 1$ and the series is divergent.
- If $|x| = 1$, then $L = 1$ and the root test gives no information. In this case, by standard limit (iii),

$$|a_n| = \left(\frac{n}{n+1} \right)^n = \frac{1}{\left(1 + \frac{1}{n}\right)^n} \rightarrow \frac{1}{e} \neq 0.$$

So for $x = 1$ the terms tend to $\frac{1}{e}$, and for $x = -1$ they alternate in sign with size approaching $\frac{1}{e}$. In both cases $a_n \not\rightarrow 0$, and the series is divergent by the n th term test.

Hence the series converges if $-1 < x < 1$ and diverges if $|x| \geq 1$.

4 Taylor and Maclaurin Series

Definition 4.1: Taylor and Maclaurin series

If f has derivatives of all orders at $x = a$, its *Taylor series* about $x = a$ is

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \cdots = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

The special case $a = 0$ is the *Maclaurin series*

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n.$$

Note:

The sum of the terms up to $(x-a)^k$ is the *Taylor polynomial of degree k* ; it approximates $f(x)$ well when x is close to a . The equality of f with its full series holds on an interval of convergence, which can be found by the ratio test. Inside that interval a power series may be differentiated or integrated term by term.

Note:

Method. Compute $f(a), f'(a), f''(a), \dots$ as far as needed, divide the n th derivative by $n!$, and attach $(x-a)^n$. Arranging the derivatives in a table keeps the work tidy.

4.1 Worked problems

Problem 4.1. [Tutorial 2, Taylor Q1] Obtain the Maclaurin series expansion, as an infinite series, of each of the following functions.

- (a) $\sin x$;
- (b) $\cos x$;
- (c) e^x ;
- (d) $\log(1+x)$;
- (e) $(1+x)^k$, where k is a real constant.

Solution: Each expansion follows the same steps: compute $f(0), f'(0), f''(0), \dots$, substitute them into the Maclaurin series

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \cdots,$$

and read off the pattern.

- (a) **Given:** $f(x) = \sin x$. The derivatives and their values at $x = 0$ are

	f	f'	f''	f'''	$f^{(4)}$	$f^{(5)}$
at x	$\sin x$	$\cos x$	$-\sin x$	$-\cos x$	$\sin x$	$\cos x$
at $x = 0$	0	1	0	-1	0	1

The derivatives repeat after every four steps, so the values at 0 cycle through 0, 1, 0, -1. Substituting,

$$\sin x = 0 + 1 \cdot x + \frac{0}{2!}x^2 + \frac{-1}{3!}x^3 + \frac{0}{4!}x^4 + \frac{1}{5!}x^5 + \cdots = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots$$

Only odd powers occur, with alternating signs:

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \quad \text{valid for all real } x.$$

(b) **Given:** $f(x) = \cos x$. The derivatives and their values at $x = 0$ are

	f	f'	f''	f'''	$f^{(4)}$	$f^{(5)}$
at x	$\cos x$	$-\sin x$	$-\cos x$	$\sin x$	$\cos x$	$-\sin x$
at $x = 0$	1	0	-1	0	1	0

The values at 0 cycle through 1, 0, -1, 0. Substituting,

$$\cos x = 1 + 0 \cdot x + \frac{-1}{2!}x^2 + \frac{0}{3!}x^3 + \frac{1}{4!}x^4 + \frac{0}{5!}x^5 + \cdots = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$$

Only even powers occur, with alternating signs:

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, \quad \text{valid for all real } x.$$

(c) **Given:** $f(x) = e^x$. Every derivative of e^x is e^x :

	f	f'	f''	f'''	$f^{(4)}$
at x	e^x	e^x	e^x	e^x	e^x
at $x = 0$	1	1	1	1	1

Substituting,

$$e^x = 1 + 1 \cdot x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \frac{1}{4!}x^4 + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad \text{valid for all real } x.$$

(d) **Given:** $f(x) = \log(1+x)$. The derivatives and their values at $x = 0$ are

	f	f'	f''	f'''	$f^{(4)}$
at x	$\log(1+x)$	$\frac{1}{1+x}$	$-\frac{1}{(1+x)^2}$	$\frac{2}{(1+x)^3}$	$-\frac{6}{(1+x)^4}$
at $x = 0$	0	1	-1	2	-6

The pattern is $f^{(n)}(x) = \frac{(-1)^{n-1}(n-1)!}{(1+x)^n}$, so $f^{(n)}(0) = (-1)^{n-1}(n-1)!$. Substituting,

$$\log(1+x) = 0 + 1 \cdot x + \frac{-1}{2!}x^2 + \frac{2}{3!}x^3 + \frac{-6}{4!}x^4 + \cdots = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots,$$

since $\frac{(n-1)!}{n!} = \frac{1}{n}$. Hence

$$\log(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n}, \quad \text{valid for } -1 < x \leq 1.$$

- (e) **Given:** $f(x) = (1+x)^k$, with k a real constant. The derivatives and their values at $x = 0$ are

	at x	at $x = 0$
f	$(1+x)^k$	1
f'	$k(1+x)^{k-1}$	k
f''	$k(k-1)(1+x)^{k-2}$	$k(k-1)$
f'''	$k(k-1)(k-2)(1+x)^{k-3}$	$k(k-1)(k-2)$
$f^{(4)}$	$k(k-1)(k-2)(k-3)(1+x)^{k-4}$	$k(k-1)(k-2)(k-3)$

Substituting,

$$(1+x)^k = 1 + kx + \frac{k(k-1)}{2!}x^2 + \frac{k(k-1)(k-2)}{3!}x^3 + \frac{k(k-1)(k-2)(k-3)}{4!}x^4 + \dots,$$

the *binomial series*, valid for $|x| < 1$. When k is a non-negative integer every term beyond x^k is zero, and the series reduces to the finite binomial theorem, valid for all x .

Problem 4.2. [Tutorial 2, Taylor Q2] Obtain the Taylor series expansion of $\log_e x$ about $x = 1$ up to the term containing $(x-1)^4$, and hence find the approximate value of $\log_e(1.1)$.

Solution: Given: $f(x) = \log x$, $a = 1$. The derivatives and their values at $x = 1$ are

	f	f'	f''	f'''	$f^{(4)}$
at x	$\log x$	$\frac{1}{x}$	$-\frac{1}{x^2}$	$\frac{2}{x^3}$	$-\frac{6}{x^4}$
at $x = 1$	0	1	-1	2	-6

Substituting into the Taylor series about $a = 1$,

$$\log x = 0 + 1(x-1) - \frac{1}{2!}(x-1)^2 + \frac{2}{3!}(x-1)^3 - \frac{6}{4!}(x-1)^4 + \dots,$$

that is,

$$\log x \approx (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \frac{(x-1)^4}{4}.$$

For $x = 1.1$, $x - 1 = 0.1$, and

$$\log(1.1) \approx 0.1 - \frac{0.01}{2} + \frac{0.001}{3} - \frac{0.0001}{4} = 0.1 - 0.005 + 0.000333 - 0.000025 = 0.0953.$$

The true value is $\log(1.1) = 0.0953102\dots$, so the approximation is correct to four decimal places.

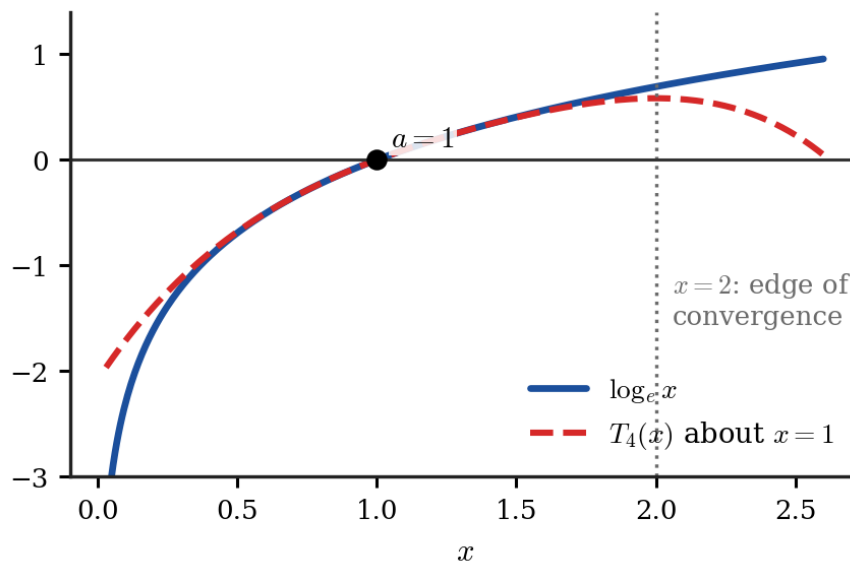


Figure 4.1: $\log x$ and its Taylor polynomial of degree 4 about $x = 1$. Near $x = 1$, and in particular at $x = 1.1$, the two are indistinguishable.

Problem 4.3. [Tutorial 2, Taylor Q4] Obtain the Taylor series expansion of $f(x) = \sin x$ about $x = \frac{\pi}{6}$ up to the term containing $\left(x - \frac{\pi}{6}\right)^5$.

Solution: Given: $f(x) = \sin x$, $a = \frac{\pi}{6}$. Here $\sin a = \frac{1}{2}$ and $\cos a = \frac{\sqrt{3}}{2}$. The derivatives cycle through $\sin x$, $\cos x$, $-\sin x$, $-\cos x$:

	f	f'	f''	f'''	$f^{(4)}$	$f^{(5)}$
at x	$\sin x$	$\cos x$	$-\sin x$	$-\cos x$	$\sin x$	$\cos x$
at $x = \frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$

Dividing each value by the factorial of its order,

$$\sin x = \frac{1}{2} + \frac{\sqrt{3}}{2} \left(x - \frac{\pi}{6}\right) - \frac{1}{4} \left(x - \frac{\pi}{6}\right)^2 - \frac{\sqrt{3}}{12} \left(x - \frac{\pi}{6}\right)^3 + \frac{1}{48} \left(x - \frac{\pi}{6}\right)^4 + \frac{\sqrt{3}}{240} \left(x - \frac{\pi}{6}\right)^5 + \dots$$

using $2! = 2$, $3! = 6$, $4! = 24$ and $5! = 120$.

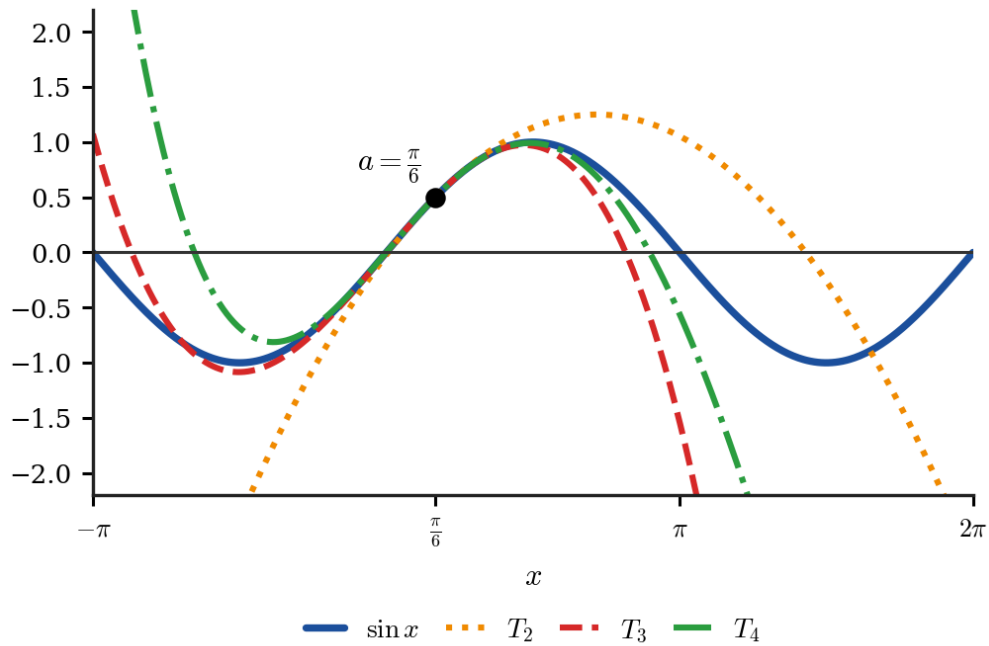


Figure 4.2: $\sin x$ and its Taylor polynomials about $a = \frac{\pi}{6}$. Each additional term keeps the polynomial close to $\sin x$ over a wider stretch around a .

Problem 4.4. [Tutorial 2, Taylor Q11] Evaluate $\int e^{-x^2} dx$ as an infinite series, using the Maclaurin series of e^{-x^2} ; write the terms of e^{-x^2} up to x^4 explicitly.

Solution: Given: the integrand e^{-x^2} . This integral cannot be expressed in terms of elementary functions, but its integrand has a simple series. Replacing x by $-x^2$ in the Maclaurin series of e^x (Problem 4.1(c)),

$$e^{-x^2} = 1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{n!},$$

valid for all real x . Integrating term by term,

$$\int e^{-x^2} dx = x - \frac{x^3}{3} + \frac{x^5}{5 \cdot 2!} - \frac{x^7}{7 \cdot 3!} + \cdots + C = x - \frac{x^3}{3} + \frac{x^5}{10} - \frac{x^7}{42} + \cdots + C,$$

and in general

$$\int e^{-x^2} dx = C + \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)n!}.$$

Using the terms of e^{-x^2} up to x^4 gives $\int e^{-x^2} dx \approx x - \frac{x^3}{3} + \frac{x^5}{10} + C$.

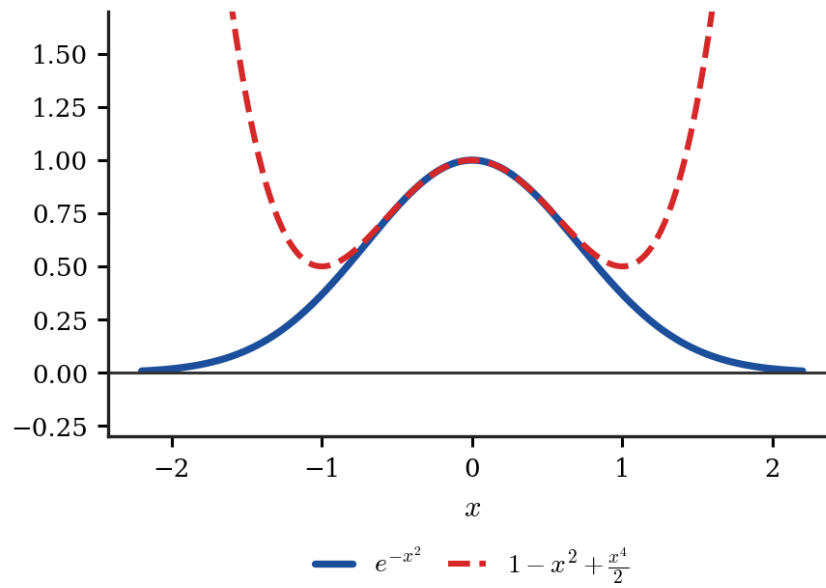


Figure 4.3: e^{-x^2} and the polynomial $1 - x^2 + \frac{x^4}{2}$ formed from its first three terms. They agree closely for $|x| < 0.6$ or so.

References

- [1] J. Stewart, D. K. Clegg and S. Watson, *Calculus: Early Transcendentals*, 9th edition, Cengage Learning, Boston, 2021.